DPhil Thesis Draft

Alessandro Lodi

Trinity College

University of Oxford

A thesis submitted for the degree of

Doctor of Philosophy

Michaelmas 2021

Contents

List of Figures

List of Abbreviations

1-D, 2-D . . . One- or two-dimensional, referring in this thesis to spatial dimensions in an image.

Otter . . . . . One of the finest of water mammals.

Hedgehog . . . Quite a nice prickly friend.

1
Introduction

Contents

1.1 Motivation

Motivation

1.2 Carbon-based Field Effect Transistor

Detail here the carbon nanotube journey.

Device type	$I_{\text{on}}/I_{\text{off}}$	SS (mV/dec)	g_{mt} (µS cm^{-1})	L_{ch} (nm)	f_T [$f_{\max}$] (GHz)	Ref.
Bottom gate	10^4	94	~ 40	9	—	[1]
	10^5	~ 85	40	15	—	[2]
	10^5	~ 85	20	20	—	[2]
	10^5	60	20	200	—	[3]
	10^5	~ 63	—	200	—	[4]
	10^4	80	50	240	—	[5]
	—	—	143	800	[1]	[6]
	10^4	—	—	1,500	1.5 – 3.0	[7]
	—	70	$12^{\dagger}$	2,000^1	—	[8]
	10^8	—	—	$< 10,000^1$	—	[9]
	—	—	- 500	—	30	[10]
Top gate	10^6	—	20^2	80	—	[11]
	2	—	~ 50	100	25	[12]
	50	—	310	100	~ 70 [80]	[13]
	—	—	0.3	210	—	[14]
	10^5	—	33.5	85,000	—	[15]

Table 1.1: Figures of merit for a few CNFET reported in literature. S: subthreshold swing; g_{m}: intrinsic transconductances; L_{ch}: channel length; μ: carriers mobility

†Normalized by the tube width.
^{1} Referred to the gated length of a nanotube.
^{2} For the n-type transistor, whereas 10 µS were measured for the p-type.

Device type	I_{on} (A)	I_{off} (A)	$I_{\text{on}}/I_{\text{off}}$	g_{mt} (µS)	SS (mV/dec)
Conventional	$1.41{*}10^{-5}$	$4.06{*}10^{-10}$	$3.4{*}10^{4}$	29	99.6
Partial S/D-doped	$1.50{*}10^{-5}$	$2.33{*}10^{-8}$	$6.4{*}10^{2}$	63.6	93.1
Schottky-Barrier	$3.26{*}10^{-6}$	$2.31{*}10^{-8}$	$1.41{*}10^{2}$	19.6	109.8
Tunnel	$5.53{*}10^{-6}$	$1.16{*}10^{-8}$	$4.76{*}10^{2}$	13	107.9

Table 1.2: Comparative simulation for the four major type of CNFETs as reported in [16]. These results are obtained by setting $V_{\text{sd}} = 0.5$ V.

1.3 Graphene Nanoribbons

The scientific interest for graphene nanoribbons started in the physics community. Although this may appear irrelevant at the first glance, we will see that this actually had an impact over the last twenty years of progress in the field. The physics community was looking for a way to open a bandgap in graphene. As mentioned in Sec. 1.1, graphene is a semi-metal, therefore unsuitable for all those semiconducting applications where the conduction paths for the charge carriers are well defined.

The first class of methods for synthesising graphene nanoribbons are methods that involve cutting or producing rectangular shapes from the graphene lattice through macroscopic methods. For this reason, they are also known as *top-down* approaches. Although they have seen an improvements in recent years, common drawbacks for this class of methods are a random distribution in the widths, a non-atomical control of the edge structures, and non-negligible impurity concentrations left after the process.

Kim and co-workers synthesized GNRs by employing the electron-beam lithography and etching technique [17]. The widths of GNRs obtained by this technique were scattered over a broad range, and most of the edges were extremely rough. An 8 nm GNR FET showed a dramatically improved on/off ratio of about 160.

Nanocutting of graphene is another promising method of GNR synthesis. In 2008, Strachan et al. used thermally activated Fe nanoparticles as chemical scissors to cut graphene sheet along a specific crystallographic direction [17, 18]. Later, Ni and other catalysts were employed as well The carbon-carbon bond breaking is driven by the combustion reaction $\mathrm{Ni} + \mathrm{C_{graphene}} + 2\,\mathrm{H_2}\uparrow \longrightarrow \mathrm{Ni/Fe} + \mathrm{CH_4}\uparrow$ catalysed by the nanoparticles, thus producing hydrocarbon compounds. One of the reason that have discouraged these synthetic way is the random width distribution obtained: this is likely to be a consequence of the change in direction when nanoparticles encounter defects or edges.

It will be interesting to discuss the $E_\mathrm{g} \propto 1/W$ relation as the model used for describing gnr is the particle in the box, but in the toy model (particle in a box) as a relation of the kind $E_\mathrm{g} \propto 1/W^2$.

1.4 Delocalised π-electron Systems

Porphyrin

2
Foundations

Contents

2.1 Charge Transport In Zero-Dimensional Objects

quantum dots transport physics

2.2 Charge Transport in One-Dimensional Objects

nanowires transport physics

2.3 Thermoelectric Phenomena in Zero-Dimensional Structures

What are the necessary changes that one has to take into account when dealing also with other extensive variable such as a temperature gradient

2.4 Density Functional Theory

What are the relevant parameters/quantities that we are interested to calculate ab initio Why we opted for a pseudopotential such as SIESTA rather than others (quantum espresso). Explain why calculating the isolated gas phase all-electron electronic structure is legit for a qualitative see how the electron density is distributed over the system.

Remember also to emphasise the difference from stricly and non-stricly localised functions used for computing the electronic properties: when the functions are stricly localised they are zero outside a given radius from the atom they are associated with and so the radius is a key quality determined parameter for the basis set

2.5 Non-Equilibrium Green Function Theory

This section is needed to explain formally the results from transiesta and TB-trans calculations

3
Instrumentations

Contents

3.1 Room Temperature DC Measurements

Describe here the probe station setup]

3.2 Dipstick DC Measurements

Describe here the dipstick used for both liquid N_2 and liquid He

3.3 Cryogenic DC Measurements

Describe here the delft box. Describe how the delft box modular structure enables to adapt two and multi-terminal DC measurements

3.4 Cryogenic AC Measurements

Describe here the delft box + lock-in setup and why it is so complicated to set it usepackage

3.5 Thermoelectric Measurements

Describe here the setup used for the thermoelectric devices and why it needs a dedicated section Describe also here from a setup point of view which complications have raised during the setup (direct lines instead of dc lines, the need to use nano-D connectors and to design the a new pcb board).

4
Fabrication

Contents

4.1 General Fabrication Procedure

The steps are well known: chip preparation, e-beam, development, metal deposition and lift-off, graphene transfer old process, graphenea graphene

4.2 Graphene Nanogaps

Here three methods have been exploited: standard feedback electroburning, gate-dependent electroburning, and litographic patterning of graphene nanogap

4.3 Chip Design

Here the designs for the different chips that have been used. Give a proper credit to Dr. Alex Gee for the fabrication of thermoelectric and the common gate ones.

4.3.1 2T Design

How convenient and 'quick' is the usage of a backgate device. List, as usual, pro and cons

4.3.2 3T Design

Why the need to use 3T over 2T

Local Gates from Waterloo

Give proper credit

Interconnected Drain and Gate

The one I made and also made with the help of Alex

4.3.3 Thermoelectric Device

Common and different features of the thermoelectric

4.4 Deposition Process and Related Phenomna

Why it has been important to try different deposition techiniques How this has been qualitatevely addressed.

5

Room Temperature Graphene Nanoribbon Single-Electron Transistor

Contents

5.1 Introduction

A brief intro why single-electron transistors with gnr are interesting A brief intro to what we are gonna do/measure in this chapter

5.2 Sample Preparation and Characterisation

How the sample it has been prepared (lot of crossreference with previous chapters) Explained why the scarsity of characterisation checks is intrinsically scarse in molecular electronics.

5.3 Electrical Transport Measurements

What type of experiments have been performed and which tricks have been used in triton to do so. What are the evolution in temperature of the zero-bias, low-bias and high bias conductance. Look at the dependence in temperature and explain possible interpretations: why it is not a Luttinger Liquid (the gnr has a bandgap) and what possible phonon modes are responsible at some point for desctructing the

5.4 Summary

6

Chemical Doping of the Bandgap for Cove Graphene Nanoribbons

Contents

6.1 Introduction

A brief intro on how why it is important to study this

6.2 Sample Preparation and Characterisation

How the sample it has been prepared (lot of crossreference with previous chapters)

Explained why the scarsity of characterisation checks i s intrinsically scarse in molecular electronics.

6.3 *Ab-initio* Experiments

What DFT and NEGF have been used and for what.

6.4 Electrical Transport Measurements

Explain how the calculations have initially suggested the comparison between the base case and the methoxy groups, as they have the largest badgap distance.

6.5 Summary

7
Thermocurrent Spectroscopy of cove Graphene Nanoribbons

Contents

7.1 Introduction

A brief intro on how why it is important to study this what has been done.

7.2 Sample Preparation and Characterisation

How the sample it has been prepared (lot of crossreference with previous chapters) Explained why the scarsity of characterisation checks i s intrinsically scarse in molecular electronics.

7.3 Electrical Transport Measurements

Explain how the calculations have initially suggested the comparison between the base case and the methoxy groups, as they have the largest badgap distance.

7.4 Summary

8

Transport Spectroscopy of butadiyne-linked Dy–Tb Double-Porphyrin Unit

Contents

8.1 Introduction

A brief intro on how why it is important to study this

8.2 Sample Preparation and Characterisation

How the sample it has been prepared (lot of crossreference with previous chapters) Explained why the scarsity of characterisation checks i s intrinsically scarse in molecular electronics.

8.3 Electrical Transport Measurements

Explain how the calculations have initially suggested the comparison between the base case and the methoxy groups, as they have the largest badgap distance.

8.4 Summary

9
Conclusions and Outlook

Contents

9.1 Conclusions

Here the final conclusions

9.2 Outlook

Appendices

Cor animalium, fundamentum eft vitæ, princeps omnium, Microcofmi Sol, a quo omnis vegetatio dependet, vigor omnis & robur emanat.

The heart of animals is the foundation of their life, the sovereign of everything within them, the sun of their microcosm, that upon which all growth depends, from which all power proceeds.

— William Harvey [19]

A

Review of Cardiac Physiology and Electrophysiology

Contents

Appendices are just like chapters. Their sections and subsections get numbered and included in the table of contents; figures and equations and tables added up, etc. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed et dui sem. Aliquam dictum et ante ut semper. Donec sollicitudin sed quam at aliquet. Sed maximus diam elementum justo auctor, eget volutpat elit eleifend. Curabitur hendrerit ligula in erat feugiat, at rutrum risus suscipit. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Integer risus nulla, facilisis eget lacinia a, pretium mattis metus. Vestibulum aliquam varius ligula nec consectetur. Maecenas ac ipsum odio. Cras ac elit consequat, eleifend ipsum sodales, euismod nunc. Nam vitae tempor enim, sit amet eleifend nisi. Etiam at erat vel neque consequat.

A.1 Anatomy

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Donec accumsan cursus neque. Pellentesque eget tempor turpis, quis malesuada dui. Proin egestas, sapien sit amet feugiat vulputate, nunc nibh mollis nunc, nec auctor turpis purus sed metus. Aenean consequat leo congue volutpat euismod. Vestibulum et vulputate nisl, at ultrices ligula. Cras pulvinar lacinia ipsum at bibendum. In ac augue ut ante mollis molestie in a arcu.

Etiam vitae quam sollicitudin, luctus tortor eu, efficitur nunc. Vestibulum maximus, ante quis consequat sagittis, augue velit luctus odio, in scelerisque arcu magna id diam. Proin et mauris congue magna auctor pretium id sit amet felis. Maecenas sit amet lorem ipsum. Proin a risus diam. Integer tempus eget est condimentum faucibus. Suspendisse sem metus, consequat vel ante eget, porttitor maximus dui. Nunc dapibus tincidunt enim, non aliquam diam vehicula sed. Proin vel felis ut quam porta tempor. Vestibulum elit mi, dictum eget augue non, volutpat imperdiet eros. Praesent ac egestas neque, et vehicula felis.

Pellentesque malesuada volutpat justo, id eleifend leo pharetra at. Pellentesque feugiat rutrum lobortis. Curabitur hendrerit erat porta massa tincidunt rutrum. Donec tincidunt facilisis luctus. Aliquam dapibus sodales consectetur. Suspendisse lacinia, ipsum sit amet elementum fermentum, nulla urna mattis erat, eu porta metus ipsum vel purus. Fusce eget sem nisl. Pellentesque dapibus, urna vitae tristique aliquam, purus leo gravida nunc, id faucibus ipsum magna aliquet ligula. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Proin sem lacus, rutrum eget efficitur sed, aliquam vel augue. Aliquam ut eros vitae sem cursus ultrices ut ornare urna. Nullam tempor porta enim, in pellentesque arcu commodo quis. Interdum et malesuada fames ac ante ipsum primis in faucibus. Curabitur maximus orci purus, ut molestie turpis pellentesque ut.

Donec lacinia tristique ultricies. Proin dignissim risus ut dolor pulvinar mollis. Proin ac turpis vitae nibh finibus ullamcorper viverra quis felis. Mauris pellentesque neque diam, id feugiat diam vestibulum vitae. In suscipit dui eu libero ultrices, et sagittis nunc blandit. Aliquam at aliquet ex. Nullam molestie pulvinar ex vitae interdum. Praesent purus nunc, gravida id est consectetur, convallis elementum nulla. Praesent ex dolor, maximus eu facilisis at, viverra eget nulla. Donec ullamcorper ante nisi. Sed volutpat diam eros. Nullam egestas neque non tortor aliquet, sed pretium velit tincidunt. Aenean condimentum, est ac vestibulum mattis, quam augue congue augue, mattis ultrices nibh libero non ante. Lorem ipsum dolor sit amet, consectetur adipiscing elit.

Aenean volutpat eros tortor, non convallis sapien blandit et. Maecenas faucibus nulla a magna posuere commodo. Nullam laoreet ante a turpis laoreet malesuada. Phasellus in varius sem. Vestibulum sagittis nibh sed tincidunt blandit. Donec aliquam accumsan odio sit amet lacinia. Integer in tellus diam. Vivamus varius massa leo, vitae ullamcorper metus pulvinar sed. Maecenas nec lorem ornare, elementum est quis, gravida massa. Suspendisse volutpat odio ex, ac ultrices leo ultrices vel. Sed sed convallis ipsum. Pellentesque euismod a nulla sed rhoncus. Sed vehicula urna vitae mi aliquet, non sodales lacus ullamcorper. Duis mattis justo turpis, id tempus est tempus eu. Curabitur vitae hendrerit ligula.

Curabitur non pretium enim, in commodo ligula. Etiam commodo eget ligula a lacinia. Vestibulum laoreet ante tellus, vel congue sapien ornare in. Donec venenatis cursus velit vitae pulvinar. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Suspendisse in metus lectus. Pellentesque gravida dolor eget finibus imperdiet. Duis id molestie tortor. Mauris laoreet faucibus facilisis. Aliquam vitae dictum massa, sit amet dignissim lacus.

Fusce eleifend tellus id ex consequat maximus. Donec ultrices ex ut turpis ornare, non molestie mi placerat. Nulla sit amet auctor nunc, sit amet euismod elit. Phasellus risus tellus, condimentum a metus et, venenatis tristique urna. Cras mattis felis eget ipsum fermentum egestas. Ut augue odio, venenatis id convallis vel, congue quis augue. Maecenas sed maximus est, posuere aliquet tortor. Ut condimentum egestas nisi eu porttitor. Ut mi turpis, posuere id lorem vel, elementum tempor arcu.

Morbi nisl arcu, venenatis non metus ac, ullamcorper scelerisque justo. Nulla et accumsan lorem. Mauris aliquet dui sit amet libero aliquet, in ornare metus porttitor. Integer ultricies urna eu consequat ultrices. Maecenas a justo id purus ultricies posuere sed et quam. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Sed eleifend risus quis aliquet gravida. Nullam ac erat porta est bibendum dictum in a dolor. Nam eget turpis viverra, vulputate lectus eget, mattis ligula. Nam at tellus eget dui lobortis sodales et ut augue. In vestibulum diam eget mi cursus, ut tincidunt nulla pellentesque.

Aliquam erat volutpat. Sed ultrices massa id ex mattis bibendum. Nunc augue magna, ornare at aliquet gravida, vehicula sed lorem. Quisque lobortis ipsum eu posuere eleifend. Duis bibendum cursus viverra. Nam venenatis elit leo, vitae feugiat quam aliquet sed. Cras velit est, tempus ac lorem sed, pharetra lobortis ipsum. Donec suscipit gravida interdum. Nunc non finibus est. Nullam turpis elit, tempus non ante.

A.2 Mechanical Cycle

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Aenean tellus est, suscipit sed facilisis quis, malesuada at ipsum. Nam tristique urna quis quam iaculis, et mattis orci pretium. Praesent euismod elit vel metus commodo ultrices. Vestibulum et tincidunt ex, in molestie ex. Donec ullamcorper sollicitudin accumsan. Etiam ac leo turpis. Duis a tortor felis. Nullam sollicitudin eu purus ac hendrerit. Nam hendrerit ligula libero, eget finibus orci bibendum a. Aenean ut ipsum magna.

Ut viverra, sapien sed accumsan blandit, nisi sem tempus tellus, at suscipit magna erat ornare nunc. Proin lacinia, nisi ut rutrum malesuada, nibh quam pellentesque nunc, sit amet consectetur purus felis ac tortor. Suspendisse lacinia ipsum eu sapien pellentesque mattis. Mauris ipsum nunc, placerat non diam vel, efficitur laoreet nunc. Sed lobortis, ipsum eget gravida facilisis, sem nulla viverra mi, in placerat eros sem viverra lacus. Aliquam porta aliquet diam vel commodo. Nulla facilisi. Duis erat libero, lobortis vel hendrerit vitae, sagittis id dui. Nulla pretium eros nec quam tincidunt, vel luctus mi aliquam. Integer imperdiet purus in est tristique venenatis. Ut pellentesque, nunc vitae iaculis ultricies, urna turpis dignissim risus, a laoreet felis magna nec erat.

Quisque sollicitudin faucibus ligula, et egestas nibh dictum sit amet. Proin eu mi a lectus congue pretium eu quis arcu. Suspendisse vehicula libero eu ipsum aliquam, vel elementum nibh mattis. Sed sed sapien vitae turpis tristique pulvinar a ut metus. Etiam semper gravida est, mollis gravida est porta ac. Proin eget tincidunt erat. Maecenas ultrices erat eget purus ultricies, ut lacinia arcu dictum. Nam et nisi sit amet ex congue mattis vel eget lorem. Aliquam erat volutpat. Pellentesque porttitor nibh vitae elementum consectetur. Aenean et est lobortis, congue sapien non, ullamcorper sapien. Ut facilisis sem non dapibus vehicula.

Mauris euismod odio dolor, sit amet gravida mauris placerat et. Curabitur nec dolor non nibh molestie lobortis dignissim non ante. Nullam rutrum lobortis ultrices. Aenean ex erat, elementum sed maximus id, posuere id quam. Proin rutrum ex elit, pretium aliquam risus finibus at. Aenean egestas orci velit, sed aliquet sapien condimentum a. Duis consequat, arcu eu viverra venenatis, dolor lorem gravida lectus, non aliquet nisi sem at augue. Donec laoreet blandit luctus. Aenean vehicula nisl vel faucibus luctus. Sed ut semper velit, vitae laoreet magna. Sed at interdum magna.

Sed iaculis faucibus odio, eu aliquam purus efficitur vel. Cras at nulla ac enim congue varius ut et nulla. Integer blandit mattis augue.

A.3 Electrical Cycle

Lorem ipsum dolor sit amet, consectetur adipiscing elit. In faucibus condimentum rhoncus. Ut dictum nisl id risus gravida lobortis. Sed vehicula mollis tellus ut varius. Fusce eget egestas dui, et commodo dui. Proin sollicitudin interdum tempus. Nullam in elit a enim fringilla bibendum. Vestibulum sodales pellentesque condimentum. Nulla facilisi. Nunc et dolor in nulla eleifend dictum at vel ligula. Aliquam ut velit non elit ullamcorper porta ac et ex. Fusce ornare magna non nunc vestibulum, eget molestie quam dictum. In interdum aliquam odio, in posuere tellus convallis quis. Curabitur non diam elit. Proin vulputate orci diam, a tincidunt ante luctus eu. Ut a viverra ligula. Curabitur pulvinar tempus tellus eget suscipit.

Aliquam posuere massa at ante dapibus congue. Curabitur ullamcorper tortor eget consectetur aliquet. Mauris tempor magna id mauris fringilla, a varius erat blandit. Nam eleifend ullamcorper placerat. Phasellus augue tortor, volutpat bibendum lorem nec, fringilla volutpat nisl. Mauris cursus urna metus, vel eleifend orci iaculis ut. Sed sit amet scelerisque massa, quis consequat dui. Donec semper sem dui, ac placerat velit egestas vel. Nulla facilisi. Quisque tellus eros, sagittis malesuada augue ut, faucibus dictum nulla. Vestibulum non dapibus erat, ut consequat libero. Ut turpis mi, dapibus commodo libero lobortis, maximus vestibulum lectus. Vestibulum sit amet sapien dapibus, tincidunt leo in, suscipit arcu. Sed in erat bibendum, laoreet eros eu, pellentesque justo. Nulla sodales purus neque, eget maximus ipsum consequat at. Maecenas a nisl sagittis, tempus ipsum sed, dictum mauris.

Suspendisse posuere odio lacus, at auctor tortor vehicula sed. Phasellus suscipit ornare enim vitae placerat. Sed viverra purus vel sapien tempor, quis iaculis erat laoreet. Aenean vel nunc vestibulum, ornare nunc ac, mollis urna. Aenean ultrices felis ipsum, ac semper est ullamcorper in. Donec in justo varius, egestas tortor ut, venenatis augue. Duis mattis, ligula quis lacinia fringilla, tellus neque accumsan ipsum, vitae tempor metus elit vel nibh. Curabitur porttitor urna nec sapien tempor, et porttitor velit malesuada.

Suspendisse aliquam nisl quis placerat vulputate. Proin dapibus ipsum ac ante sagittis, volutpat auctor sem dapibus. Nam in facilisis odio. Integer ante mauris, eleifend et pulvinar in, venenatis quis ligula. Phasellus posuere sollicitudin tortor eget euismod. Maecenas mollis tortor eget justo vulputate sagittis. Etiam hendrerit massa quis ex molestie sodales. Quisque facilisis erat lacus, id convallis sem suscipit bibendum. Integer dui urna, pharetra sed porta sed, bibendum ut odio. Donec placerat at lectus egestas consequat. Sed id rhoncus est, vitae vulputate sapien. Fusce tempus quam lorem, id ornare turpis sodales sed. Integer aliquet urna eget condimentum consequat. Vestibulum quis dui vel ligula posuere luctus id nec turpis.

Nam vitae placerat lacus. Mauris scelerisque interdum volutpat. Nunc aliquet tristique enim, sit amet molestie felis ullamcorper vitae. Nullam sollicitudin orci

orci, in condimentum tellus consectetur in. Nam id justo justo. Fusce eget finibus est. Proin id tortor nec quam cursus vehicula. Aliquam ultrices eros eros, a tincidunt elit eleifend auctor.

Nullam consectetur dapibus ligula sit amet efficitur. Nunc non posuere sapien. Vivamus dui nisl, aliquam id ipsum non, pulvinar ornare neque. Nunc rhoncus pretium congue. Fusce id laoreet enim. Cras sed massa in eros bibendum auctor in nec sem. Nam commodo, velit id porta consequat, mi arcu gravida lorem, ut aliquam elit ante quis dui. Quisque in massa sed nibh blandit dictum.

Vestibulum molestie consectetur porttitor. Donec tincidunt vel orci at pharetra. Nullam id felis sit amet nulla tempus lacinia. Integer egestas ullamcorper massa, ut ultricies diam congue sit amet. Cras sit amet velit at nibh vehicula finibus a et lorem. Cras odio metus, venenatis ut ultrices non, ornare ac orci. Morbi et nulla dui. Mauris dictum molestie nibh, eu efficitur lorem accumsan quis.

A.4 Cellular Electromechanical Coupling

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nullam vitae consectetur metus, ac maximus ex. Quisque vitae ex eu lectus ultricies consequat vel non lorem. Etiam odio ipsum, tempus ut lobortis in, molestie ac leo. Vivamus mollis feugiat bibendum. Vestibulum eget venenatis quam. Aenean faucibus, massa sed ullamcorper porta, arcu nunc iaculis velit, quis consectetur purus neque placerat nibh. Vestibulum elit nunc, dignissim vulputate venenatis et, sodales non massa. Proin leo ligula, vehicula eu aliquam varius, posuere a dolor. Donec iaculis auctor neque, sit amet gravida libero porta vel. Vivamus consequat elementum lacus, at bibendum mauris egestas nec. Fusce fermentum diam eu dolor ornare, vitae vestibulum leo interdum. Morbi luctus libero quis dictum laoreet. Etiam semper porta ante, vel ullamcorper enim sodales quis.

Nullam eu nisi faucibus, fermentum ex auctor, tempor arcu. Phasellus condimentum erat mi, condimentum malesuada ligula congue venenatis. Nullam gravida imperdiet urna quis cursus. Ut tempus nec purus eget posuere. Cras non nulla sit amet justo aliquet pellentesque nec sed eros. Nam aliquam nisl urna, in placerat magna gravida venenatis. Donec interdum vel magna ullamcorper molestie. Nunc felis neque, rhoncus fringilla faucibus sit amet, ultrices sed magna. Maecenas malesuada hendrerit diam in ultrices. Nam libero urna, volutpat ut auctor eget, interdum sed odio. Vestibulum suscipit mauris nec augue ornare, ut eleifend nulla gravida. Proin imperdiet, mauris quis consectetur porta, leo dui convallis leo, id lobortis massa diam eu libero. Aenean hendrerit vel ante aliquam venenatis. Pellentesque bibendum pretium odio, ut sagittis lectus feugiat a. Donec porttitor vulputate lacus.

Nunc volutpat efficitur lacus in aliquet. Nullam non iaculis diam, at ultrices diam. Proin vehicula vulputate cursus. Morbi tempus sapien id urna lobortis interdum. Maecenas elementum sagittis elementum. Donec at sodales velit, a posuere tortor. Nulla id hendrerit tortor. Sed semper velit in magna sagittis pulvinar. Nulla nec arcu molestie, ultricies sapien sit amet, sollicitudin nisi. Donec nisi massa, suscipit ut dignissim quis, lacinia id leo.

Suspendisse ut mi metus. Morbi tincidunt ligula in porttitor consectetur. Integer eu urna urna. Suspendisse potenti. Mauris sit amet felis eu diam auctor ullamcorper. Morbi in porta nisi. Nam ante tortor, venenatis vitae tempor sed, sagittis vitae velit. In semper orci sit amet nisi ullamcorper varius. Aenean dignissim ultrices imperdiet. Maecenas lacinia enim id neque porttitor iaculis. Curabitur laoreet ante ut urna dignissim, id sollicitudin metus consectetur. Aenean massa ipsum, auctor vel ante vel, blandit dignissim libero. Fusce interdum ac magna et interdum.

The first kind of intellectual and artistic personality belongs to the hedgehogs, the second to the foxes ...

— Sir Isaiah Berlin [20]

References

[1] Aaron D Franklin et al. "Sub-10 nm carbon nanotube transistor". In: *Nano letters* 12.2 (2012), pp. 758–762.

[2] Aaron D Franklin and Zhihong Chen. "Length scaling of carbon nanotube transistors". In: *Nature nanotechnology* 5.12 (2010), p. 858.

[3] Ralf Thomas Weitz et al. "High-performance carbon nanotube field effect transistors with a thin gate dielectric based on a self-assembled monolayer". In: *Nano letters* 7.1 (2007), pp. 22–27.

[4] Yu-Ming Lin et al. "High-performance carbon nanotube field-effect transistor with tunable polarities". In: *IEEE Transactions on Nanotechnology* 4.5 (2005), pp. 481–489.

[5] Ali Javey et al. "High-field quasiballistic transport in short carbon nanotubes". In: *Physical Review Letters* 92.10 (2004), p. 106804.

[6] N Chimot et al. "Gigahertz frequency flexible carbon nanotube transistors". In: *Applied Physics Letters* 91.15 (2007), p. 153111.

[7] Michael Schröter et al. "CNTFET-based RF electronics—State-of-the-art and future prospects". In: *2016 IEEE 16th Topical Meeting on Silicon Monolithic Integrated Circuits in RF Systems (SiRF)*. IEEE. 2016, pp. 97–100.

[8] Ali Javey et al. "High-κ dielectrics for advanced carbon-nanotube transistors and logic gates". In: *Nature materials* 1.4 (2002), p. 241.

[9] Vladimir Derenskyi et al. "Carbon nanotube network ambipolar field-effect transistors with 108 on/off ratio". In: *Advanced Materials* 26.34 (2014), pp. 5969–5975.

[10] A Le Louarn et al. "Intrinsic current gain cutoff frequency of 30 GHz with carbon nanotube transistors". In: *Applied physics letters* 90.23 (2007), p. 233108.

[11] Ali Javey et al. "High performance n-type carbon nanotube field-effect transistors with chemically doped contacts". In: *Nano letters* 5.2 (2005), pp. 345–348.

[12] Yuchi Che et al. "T-gate aligned nanotube radio frequency transistors and circuits with superior performance". In: *ACS nano* 7.5 (2013), pp. 4343–4350.

[13] Yu Cao et al. "Radio frequency transistors using aligned semiconducting carbon nanotubes with current-gain cutoff frequency and maximum oscillation frequency simultaneously greater than 70 GHz". In: *ACS nano* 10.7 (2016), pp. 6782–6790.

[14] Fumiyuki Nihey et al. "A top-gate carbon-nanotube field-effect transistor with a titanium-dioxide insulator". In: *Japanese journal of applied physics* 41.10A (2002), p. L1049.

[15] Pak Heng Lau et al. "Fully printed, high performance carbon nanotube thin-film transistors on flexible substrates". In: *Nano letters* 13.8 (2013), pp. 3864–3869.

[16] Amandeep Singh, Mamta Khosla, and Balwinder Raj. "Comparative analysis of carbon nanotube field effect transistors". In: *2015 IEEE 4th Global Conference on Consumer Electronics (GCCE)*. IEEE. 2015, pp. 552–555.

[17] Liang Ma, Jinlan Wang, and Feng Ding. "Recent Progress and Challenges in Graphene Nanoribbon Synthesis". In: *ChemPhysChem* 14.1 (2013), pp. 47–54.

[18] Sujit S. Datta et al. "Crystallographic Etching of Few-Layer Graphene". In: *Nano Letters* 8.7 (2008), pp. 1912–1915. URL: http://arxiv.org/pdf/0806.3965v1.pdf.

[19] William Harvey. *Exercitatio Anatomica de Motu Cordis et Sanguinis in Animalibus*. Latin. 1st. Frankfort: Sumptibus Guilielmi Fitzeri, 1628.

[20] Isaiah Berlin. *The Hedgehog and the Fox: An Essay on Tolstoy's View of History*. English. Ed. by Henry Hardy. 2nd. Princeton University Press, June 2013.